Heist Season

HackerRank

A crew of n members plans a season of heists chosen from N available heists. Each heist i (from 1 to N) requires g[i] members and yields a loot worth p[i]. Each crew member can work on at most one heist, meaning the total number of members used across all chosen heists must not exceed n.

There are two special mechanics:

1. The Fence's Cut: Loot must be fenced, and the fence charges a flat fee F per chosen heist. If the chosen heists yield a total loot of P and there are K chosen heists, the crew's take-home profit is P − K × F. The crew's take-home profit must be at least minTake.
2. The Inside Man: The crew has exactly one insider who can be assigned to at most one chosen heist. If the insider is assigned to heist i, that heist requires ceil(g[i] / 2) members instead of g[i] (where ceil(x) is the smallest integer greater than or equal to x). The insider is not a crew member, so assigning him does not consume any of the n crew members. Using the insider is optional.

A plan consists of selecting a subset of heists, and optionally choosing at most one heist from that subset to assign the insider to. Two plans are considered different if they select different subsets of heists, or if they select the same subset but assign the insider to different heists (or one assigns the insider and the other does not).

A plan is valid if:

• The total members required (after applying the halving for the insider, if used) does not exceed n.

• The crew's take-home profit is at least minTake. Count the number of valid plans. Since the number of plans can be large, return the count modulo $10^9 + 7$.

Input Format

The first line contains a integer, numHeists, denoting the number of available heists.

The second line contains a integer, crewSize, denoting the total number of crew members available.

The third line contains a integer, minTake, denoting the minimum take-home profit required.

The fourth line contains a integer, fenceFee, denoting the flat fee charged by the fence per heist

Each line i of the numHeists subsequent lines (where 0 ≤ i < numHeists) contains a integer, memberRequirements[i].

Each line i of the numHeists subsequent lines (where 0 ≤ i < numHeists) contains a integer, lootValues[i].

Constraints

1 <= numHeists <= 100

1 <= crewSize <= 100

0 <= minTake <= 100

0 <= fenceFee <= 100

1 <= memberRequirements[i] <= 100

0 <= lootValues[i] <= 100

Output Format

The output contains an integer value

Sample Input 0

```
1
1
0
0
1
0
```

Sample Output 0

```
3
```

Explanation 0

The three valid plans are choosing no heists, choosing the single heist without the insider, or choosing the single heist with the insider, all of which satisfy the crew size of 1 and minimum profit of 0.

Sample Input 1

```
5
10
100
50
2
2
2
2
2
100
100
100
100
100
```

Sample Output 1

```
101
```

Explanation 1

To meet the minimum profit of 100 with a fee of 50 per heist, at least 2 heists must be chosen. Counting all valid subsets of 2 to 5 heists, along with the optional insider assignments, results in 101 valid plans.